

බස්නාහිර පළාත් අධ්‍යාපන දෙපාර්තමේන්තුව மேல் மாகாணக் கல்வித் திணைக்களம் Department of Education - Western Province			
වර්ෂ අවසාන ඇගයීම ஆண்டிறுதி மதிப்பீடு - 2018 Year End Evaluation			
ශ්‍රේණිය } තරම } 10 Grade }	විෂය } பாடம் } Science Subject }	පත්‍රය } வினாத்தாள் } I Paper }	කාලය } காலம் } 01 Hour Time }

- (i) Answer all the questions.
- (ii) Select the most correct or most appropriate answer from the given answers (i), (ii), (iii), (iv) from question no. 01 to 40. Put (X) mark on the relevant answer in front of the question number in the answer sheet provided.
- (iii) 40 marks are allocated for this paper.

- (01) A group of cells which is adapted to perform a specific function in an organism is called as,
- (1) organ (2) organelle (3) system (4) tissue
- (02) The metal which doesn't react with cold water but reacts with hot water is,
- (1) copper (2) potassium (3) calcium (4) magnesium
- (03) What is the equipment which is always connected to the circuit parallelly?
- (1) Ammeter (2) Galvanometer
(3) Voltmeter (4) rheostat
- (04) Deficiency of which of the following vitamins leads to "beri - beri"?
- (1) Vitamin A (2) Vitamin B (3) Vitamin C (4) Vitamin K
- (05) Which the following has an ionic lattice in its structure?
- (1) Diamond (2) Graphite
(3) Sodium chloride (4) Silicon dioxide
- (06) Select the response which only includes vector quantities
- (1) Distance, displacement (2) velocity, displacement
(3) acceleration, speed (4) force, speed
- (07) Which of the following carbohydrates is a disaccharide?
- (1) Glucose (2) Galactose (3) Maltose (4) Cellulose
- (08) Which of the following is an amorphous form of carbon?
- (1) coal (2) graphite (3) diamond (4) fullerine

(09) A 30 N force was applied to an object with 10 kg mass in the same direction of movement. What is the acceleration of the object?

- (1) 0.33 ms^{-2} (2) 3 ms^{-2} (3) 30 ms^{-2} (4) 300 ms^{-2}

(10) What is the most suitable method to obtain large number of plants identical to mother plant?

- (1) grafting (2) rooting of stem cuttings
(3) sowing the seed paddy (4) tissue culture

(11) The energy transformation of a stone when throwing it using a catapult is,

- (1) kinetic energy of the rubber band $\rightarrow$ kinetic energy of the stone
(2) gravitational potential of the rubber band $\rightarrow$ kinetic energy of the stone
(3) elastic potential energy of the rubber band $\rightarrow$ kinetic energy of the stone
(4) gravitational potential energy of the rubber band $\rightarrow$ kinetic energy of the stone

(12) The electronic configuration of element A is 2, 8, 6. number of protons and electrons present in A^{2+} is given respectively as

- (1) 16, 16 (2) 16, 18 (3) 18, 16 (4) 34, 18

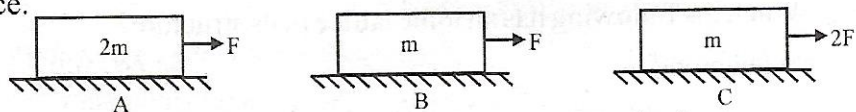
(13) Select the pair of animals with four chambers in the heart

- (1) Rabbit, Salamander (2) Tilapia, Gecko
(3) Ostrich, Whale (4) Python, Squirrel

(14) An element has four shells with electrons. It has 2 electrons in its outermost shell. The period, group and valency of that element given respectively as,

- (1) 4, IV, 2 (2) 4, II, 2 (3) 2, IV, 2 (4) 2, II, 4

(15) Diagram shows, the three instances of applying unbalanced forces on objects A, B and C placed on same surface.



The instances of highest and lowest acceleration in objects given respectively as

- (1) A and B (2) B and C (3) C and A (4) C and B

(16) What is the name given to the plant in which both staminate flowers and pistillate flowers are produced?

- (1) Monoecious plants (2) Dioecious plants
(3) Dicotomous plants (4) staminate plants

(17) Which of the following denotes a stable ion and a stable atom respectively?

- (1) Na^+ and Mg (2) Ca^{2+} and Cl (3) Na^+ and O (4) Cl^- and Ne

(18) Which of the following is not an instance where atmospheric pressure is used?

- (1) Drinking with the use of a straw
- (2) fixing an object using a rubber sucker
- (3) Removing water from a fish tank using siphon method
- (4) Lifting an object using a hydraulic jak

(19) Gene composition of an organism for a particular characteristic is known as

- (1) phenotype
- (2) genotype
- (3) linked genes
- (4) Heterozygous genes

(20) When moving from left to right along a period, in the periodic table

- (a) - First ionization energy decreases
- (b) - Electro negativity increases
- (c) - The acidity of the oxide increases

Correct statement is/are

- (1) a only
- (3) b and c only

(2) b only

(4) a and c only

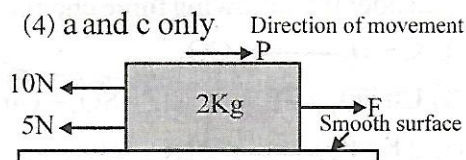
(21) What is the force 'F' which needs to be applied to move the object towards the direction P, with an acceleration of 4 ms^{-2}

(1) 10 N

(2) 15 N

(3) 5 N

(4) 23 N



(22) Few statements regarding the cellular organelle shown by diagram are given below.

- (a) It can be only viewed through an electron microscope
- (b) It is surrounded by two membranes
- (c) It carries out photosynthesis.

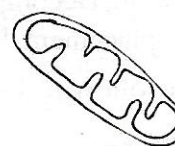
Correct statement is/are

(1) only a

(2) only b

(3) b and c only

(4) a, b only



(23) An ion is represented as $[x^-]$. If the element X is in the 3rd period of the periodic table, its electronic configuration is written as.

(1) 2, 7

(2) 2, 8, 1

(3) 2, 8, 7

(4) 2, 8, 6

(24) The following figures show three metallic wires made up of same metal. The explanation given by three students regarding x, y, z metal wires are as follows

(a) x has higher resistance than y

(b) z has less resistance than x

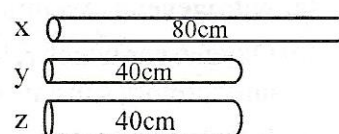
(c) Resistance of z and y are equal. The correct statements are

(1) a and b

(2) b and c

(3) a and c

(4) c only.



(25) Select the response which only include the phases taking place in ovary during menstrual cycle

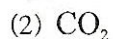
(1) Menstrual phase, Proliferation phase

(2) Follicuton phase, Luteal phase

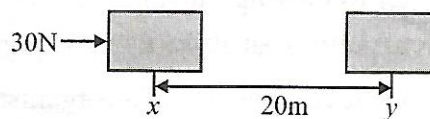
(3) Follicuton phase, secretory phase

(4) Proliferation phase, Secretory phase

(26) Out of the following compounds, the one which contain octate of electrons in last shell of each atom is



(27) An object is made to move from x to y by a student as shown in the diagram within 2 minutes



(1) $\frac{30}{2 \times 60} \text{ W}$

(2) $\frac{30 \times 20}{2} \text{ W}$

(3) $\frac{30 \times 20}{2 \times 60} \text{ W}$

(4) $\frac{30 \times 10}{4 \times 60} \text{ W}$

(28) The exact locations where fertilization and implantation should take place in female reproductive system given respectively as

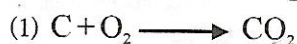
(1) ovary, upper end of the fallopian tube

(2) uterus, upper end of the fallopian tube

(3) Upper end of the fallopian tube, uterine wall

(4) ovary, vagina

(29) Consider the following three chemical reactions



The type of above reactions given respectively as

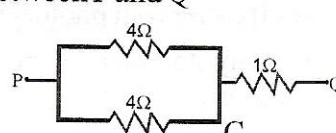
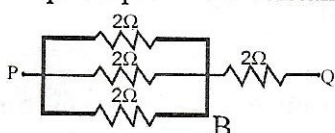
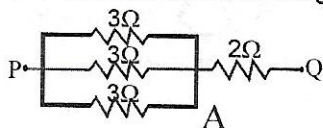
(1) Decomposition, Combination, Single displacement

(2) Combination, Single displacement, Decomposition

(3) Combination, Decomposition, Single displacement

(4) Single displacement, Decomposition, Combination

(30) Which of the following has the equal equivalent resistance between P and Q



(1) A, B only

(2) B, C only

(3) A, C only

(4) all A, B, C

(31) Following statements describe excretion of organisms

(a) Nitrogenous excretion mainly takes place in kidneys

(b) Oxygen gas which released through stomata into the atmosphere is considered as an excretory substance.

(c) Exhaled air contain excretory substance

Correct statement are

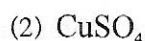
(1) a, b only

(2) b, c only

(3) a, c only

(4) a, b, c all

(32) Aqueous solution of which of the following substance produce a green colour?



(33) An object starting at rest took 5 s to reach the ground under gravitational acceleration. Find the displacement of the object during above time duration ($g = 10\text{ms}^{-2}$)

(1) 50m

(2) 100m

(3) 150m

(4) 125m

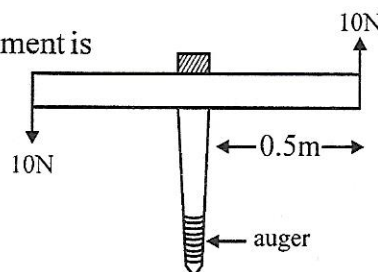
- (34) What is the advantage of consuming food with fibre?
- (1) prevent constipation
 - (2) regulating the glucose level in blood
 - (3) reducing the risk of colon cancer
 - (4) all above

- (35) What is the total numbers of atoms in 5 molecular moles?

- (1) $3 \times 6.022 \times 10^{23}$
- (2) $5 \times 6.022 \times 10^{23}$
- (3) $8 \times 6.022 \times 10^{23}$
- (4) $3 \times 5 \times 6.022 \times 10^{23}$

- (36) The moment of couple of forces exerted on the above equipment is

- (1) $10\text{N} \times 1\text{m}$
- (2) $10 \times 10\text{N} \times 0.5\text{m}$
- (3) $\frac{10}{0.5} \text{ Nm}$
- (4) $10\text{N} \times 0.5\text{m}$

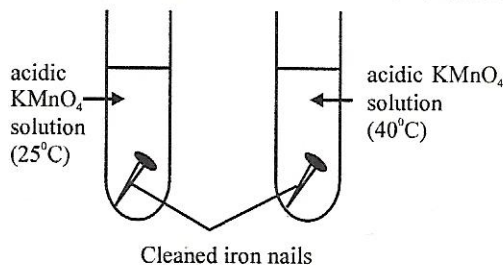


- (37) Which of the following is not an application of genetic engineering?

- (1) Production of crops resistant to weedicides
- (2) Production of tomato varieties which can withstand cold climate
- (3) For criminal investigations
- (4) To remove bladder stones in patients

- (38) The apparatus given below is used to study the rate of reaction. The factor considered in the above experiment is

- (1) Effect of concentration of reactants
- (2) Effect of catalysts
- (3) Effect of temperature
- (4) Effect of physical nature



- (39) The following diagrams show how four object A, B, C, D are balanced on a light uniform rod



Select the response which shows the descending order of weights of objects A, B, C, D

- (1) DCBA
- (2) CDAB
- (3) ABCD
- (4) BCDA

- (40) Which of the following cannot be explained by the reactivity of metal in activity series?

- (1) It is difficult to extract highly reactive metals
- (2) Gold jewellery can be kept for a long time without loss of its lustre
- (3) Aluminium does not corrode
- (4) Metals like platinum and silver exist in their native form in nature